

ECE 332: Engineering Electromagnetics II

Lecture 16

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Quick Review

Half-Wave Dipole Antenna

- Length:

$$\ell = \lambda/2 \quad (1)$$

- Current:

$$i(t) = I_0 \cos(kz) \cos(\omega t) \quad (2)$$

$$\tilde{I}(z) = I_0 \cos(kz) \quad (3)$$

where

$$-\lambda/4 \leq z \leq \lambda/4 \quad (4)$$

and the **wavenumber**

$$k = 2\pi/\lambda. \quad (5)$$

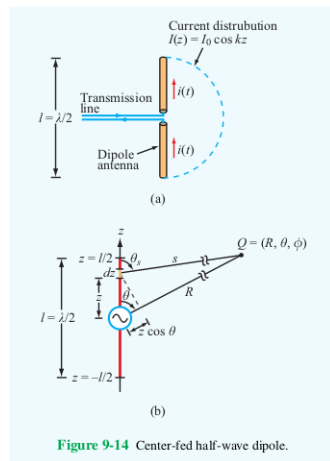


Figure 9-14 Center-fed half-wave dipole.

Quick Review (Continued)

- **Phasors of the Electric and Magnetic Fields:**

In the far field,

$$\tilde{E}_\theta = j60I_0 \left(\frac{e^{-jkR}}{R} \right) \left(\frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right) \quad (6)$$

$$\tilde{H}_\phi = \tilde{E}_\theta / \eta_0 \quad (7)$$

with $\tilde{E}_R = \tilde{E}_\phi = \tilde{H}_R = \tilde{H}_\theta = 0$.

- **Average Power Density:**

$$S(R, \theta) = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta} \right) \quad (8)$$

The maximum average power density is

$$S_{\max} = S_0 = \frac{15I_0^2}{\pi R^2}. \quad (9)$$

Quick Review (Continued)

- **Normalized Radiation Intensity:**

$$F(\theta, \phi) = F(\theta) = \frac{S(R, \theta)}{S_0} = \left(\frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right)^2 \quad (10)$$

- **Total Radiated Power:**

$$P_{\text{rad}} = 36.6 I_0^2 \text{ (W)} \quad (11)$$

- **Directivity:**

$$D = 1.64, \quad D \text{ (dB)} = 2.15 \text{ dB} \quad (12)$$

- **Radiation Resistance:**

$$R_{\text{rad}} = 73.2 \text{ } (\Omega) \quad (13)$$

- **Input Impedance:**

$$Z_{\text{in}} \approx R_{\text{rad}} + jX_{\text{in}}, \quad (14)$$

where $X_{\text{in}} = 0$ when the antenna is shortened slightly.

Quick Review (Continued)

Quarter-Wave Monopole Antenna

This antenna has **length**

$$\ell = \lambda/4 \quad (15)$$

and is positioned **above a conducting ground plane**.

- **EM Field Phasors:**

$$\begin{aligned} \tilde{E}_\theta &= j60I_0 \left(\frac{e^{-jkR}}{R} \right) \left(\frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \right), \\ \tilde{H}_\phi &= \tilde{E}_\theta / \eta_0, \end{aligned} \quad (16)$$

and $\tilde{E}_R = \tilde{E}_\phi = \tilde{H}_R = \tilde{H}_\theta = 0$ for

$$0 \leq \theta \leq \pi/2. \quad (17)$$

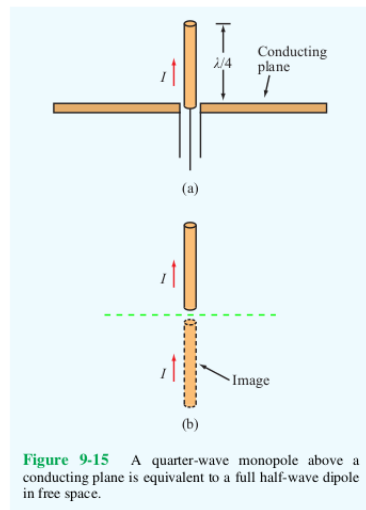


Figure 9-15 A quarter-wave monopole above a conducting plane is equivalent to a full half-wave dipole in free space.

Quick Review (Continued)

- **Normalized Radiation Intensity:**

$$F(\theta) = \left(\frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right)^2, \quad (18)$$

- **Total Radiated Power:**

$$P_{\text{rad}} = \frac{1}{2}(36.6 I_0^2) = 18.3 I_0^2 \text{ (W)}, \quad (19)$$

- **Radiation Resistance:**

$$R_{\text{rad}} = \frac{1}{2}(73.2) = 36.6 \text{ } (\Omega). \quad (20)$$

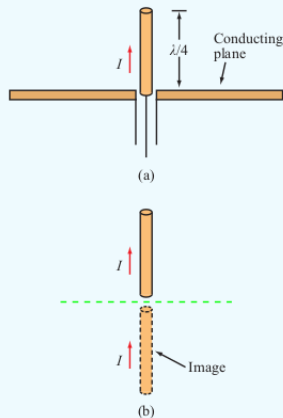


Figure 9-15 A quarter-wave monopole above a conducting plane is equivalent to a full half-wave dipole in free space.

Quick Review (Continued)

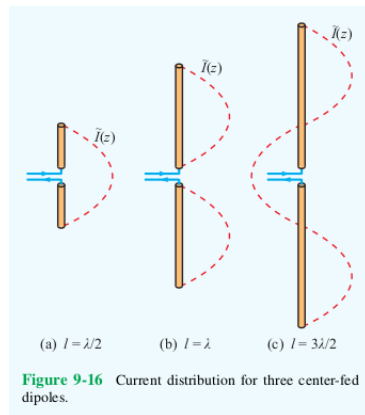
Arbitrary-Length Dipole Antennas

For a **center-fed dipole**, the currents through the two halves are symmetrical and 0 at the ends $z = -\ell/2, 0, \ell/2$:

$$\tilde{I}(z) = I_0 \begin{cases} \sin(k(\frac{\ell}{2} - z)), & 0 \leq z \leq \frac{\ell}{2} \\ \sin(k(\frac{\ell}{2} + z)), & -\frac{\ell}{2} \leq z < 0 \end{cases} \quad (21)$$

where I_0 is the **constant amplitude** and the **wavenumber** is

$$k = 2\pi/\lambda. \quad (22)$$



Quick Review (Continued)

The spherical coordinates components of the **phasors of the EM fields** are

$$\tilde{E}_\theta = j60I_0 \left(\frac{e^{-jkR}}{R} \right) \left(\frac{\cos\left(\frac{k\ell}{2} \cos\theta\right) - \cos\left(\frac{k\ell}{2}\right)}{\sin\theta} \right), \quad \tilde{H}_\phi = \tilde{E}_\theta / \eta_0, \quad (23)$$

and $\tilde{E}_R = \tilde{E}_\phi = \tilde{H}_R = \tilde{H}_\theta = 0$.

The **average power density** is

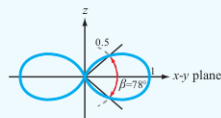
$$S(R, \theta) = \frac{|\tilde{E}_\theta|^2}{2\eta_0} = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos\left(\frac{\pi\ell}{\lambda} \cos\theta\right) - \cos\left(\frac{\pi\ell}{\lambda}\right)}{\sin\theta} \right)^2. \quad (24)$$

Quick Review (Continued)

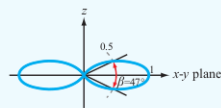
The **antenna patterns** when

$$\ell = \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2} \quad (25)$$

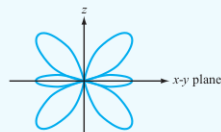
are shown in the adjacent figure. There is also a module accessible through the companion website that allows you to interactively explore the characteristics of this type of antenna.



(a) $l = \lambda/2$



(b) $l = \lambda$



(c) $l = 3\lambda/2$

Figure 9-17 Radiation patterns of dipoles with lengths of $\lambda/2$, λ , and $3\lambda/2$.

Quick Review (Continued)

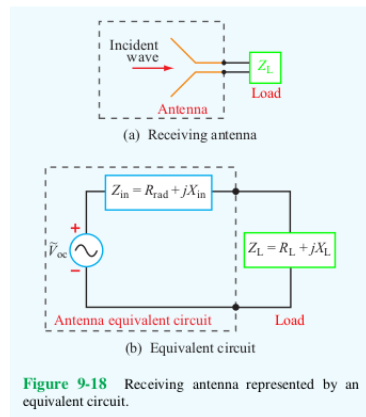
Quantities for characterizing an antenna as an EM wave receiver:

Suppose

- S_i (W/m^2) is the power density of the incident EM wave
- and the antenna collects an **intercepted power** P_{int} (W).

Then the **effective area** (also called **effective aperture** or **receiving cross section**) of the antenna is defined as

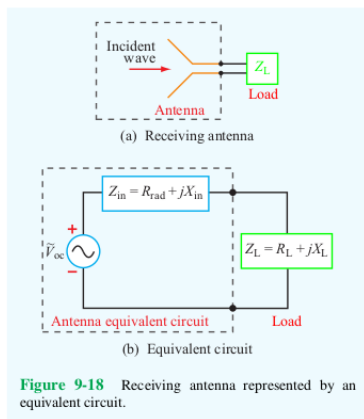
$$A_e = \frac{P_{\text{int}}}{S_i} \text{ (m}^2\text{)}. \quad (26)$$



Quick Review (Continued)

The **equivalent circuit** in the figure represents the transfer of power through the antenna to the load. Here,

- $\tilde{V}_{oc}$ is the **open-circuit voltage** that the incident wave induces at the antenna terminals,
- $Z_{in} = R_{in} + jX_{in}$ is the **input impedance** of the antenna,
- and $Z_L = R_L + jX_L$ is the **impedance of the load**.



Quick Review (Continued)

If

- $R_{\text{loss}} \ll R_{\text{rad}}$
- and Z_L and Z_{in} are chosen to make

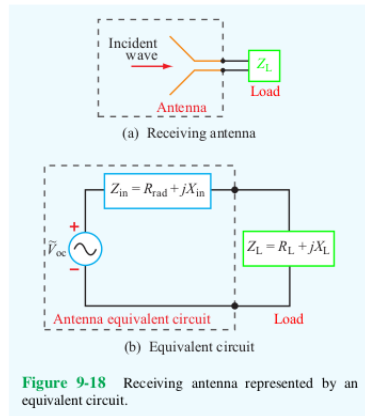
$$Z_L = Z_{\text{in}}^*, \quad (27)$$

then

- $Z_{\text{in}} = R_{\text{rad}} + jX_{\text{in}}$,
- the **overall impedance** is $2R_{\text{rad}}$,
- and the **maximum power**, which is **all of the intercepted power**,

$$P_L = P_{\text{int}} = \frac{|\tilde{V}_{\text{oc}}|^2}{8R_{\text{rad}}} \quad (28)$$

is transferred to the load.



Quick Review (Continued)

Let $\tilde{E}_i$ be the component of the incident wave **parallel to the polarization direction** of the antenna. Then the **power density of the wave** is

$$S_i = \frac{|\tilde{E}_i|^2}{240\pi} \quad (29)$$

and the **effective area of the antenna** is

$$A_e = \frac{30\pi |\tilde{V}_{oc}|^2}{R_{rad} |\tilde{E}_i|^2}. \quad (30)$$

For any antenna,

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}. \quad (31)$$

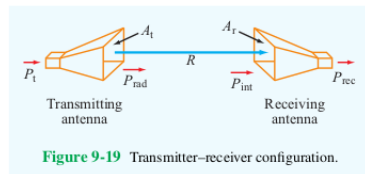
In particular, for a **Hertzian dipole**, $D = \frac{3}{2}$ (See Example 9-2.) and

$$A_e = \frac{3\lambda^2}{8\pi} \text{ (m}^2\text{)}. \quad (32)$$

Quick Review (Continued)

Suppose

- **two antennas** are oriented with the **peaks of their antenna patterns aligned**,
- one antenna **transmits**,
- the other antenna **receives**,
- and that the **transmission lines** connected to each antenna are **impedance matched** to the antennas.



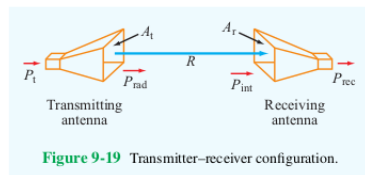
Quick Review (Continued)

For the **transmitting antenna**, let

- A_t be the **effective area**,
- ξ_t be the **radiation efficiency**,
- and P_t be the **power applied from the transmitter**.

For the **receiving antenna**, let

- A_r be the **effective area**,
- ξ_r be the **radiation efficiency**,
- and P_{rec} be the **power received at the receiver**.



Quick Review (Continued)

The **Friis transmission formula** determines the **power transfer ratio**

$$\frac{P_{\text{rec}}}{P_{\text{t}}} = \frac{\xi_{\text{t}}\xi_{\text{r}}A_{\text{t}}A_{\text{r}}}{\lambda^2 R^2} = G_{\text{t}}G_{\text{r}}\left(\frac{\lambda}{4\pi R}\right)^2. \quad (33)$$

More generally, if the **peaks of the antenna patterns are not aligned**,

$$\frac{P_{\text{rec}}}{P_{\text{t}}} = G_{\text{t}}G_{\text{r}}\left(\frac{\lambda}{4\pi R}\right)^2 F_{\text{t}}(\theta_{\text{t}}, \phi_{\text{t}})F_{\text{r}}(\theta_{\text{r}}, \phi_{\text{r}}). \quad (34)$$

Here,

- the angles $(\theta_{\text{t}}, \phi_{\text{t}})$ specify the direction of the receiving antenna relative to the antenna pattern of the transmitting antenna
- and the angles $(\theta_{\text{r}}, \phi_{\text{r}})$ specify the direction of the transmitting antenna relative to the antenna pattern of the receiving antenna.

Radiation from Large-Aperture Antennas

An **ideal wire antenna**, i.e. current source antenna, generates electromagnetic waves using currents **along a line in space** whereas an **aperture antenna** generates electromagnetic waves using **an electric field distributed across its aperture**.

Example: A Horn Antenna connected to a Coaxial Transmission Line

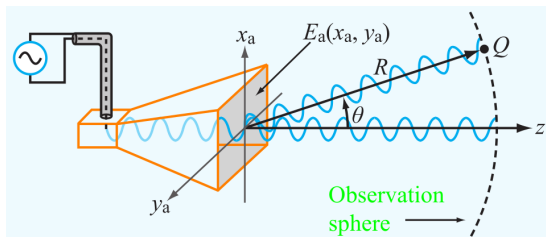


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

The horn acts as a **waveguide** that is fed by the **monopole antenna** consisting of the **protruding end of the transmission line**.

Radiation from Large-Aperture Antennas (Continued)

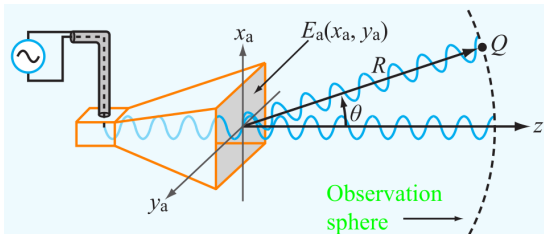


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

- **Geometry** determines the electromagnetic fields **within the horn**.
- **The aperture is the boundary** between
 - the region where the wave is guided
 - and the unbounded medium.
- **Each point** on the aperture can be thought of as an **isotropic antenna** radiating spherical waves.

Radiation from Large-Aperture Antennas (Continued)

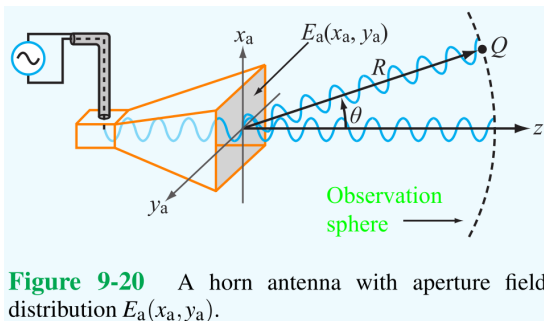


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

- The waves from each point on the aperture are **superposed** to determine the **overall wave emitted**.
- There are **two approaches** to analyzing this situation.
 - a **scalar theory** associated with Kirchhoff,
 - a **vector theory** derived from Maxwell's equations.

Radiation from Large-Aperture Antennas (Continued)

We'll explore the **scalar theory**, which is appropriate for antennas that have apertures with principal dimensions that are **at least several wavelengths long**. (Therefore, these are “large-aperture” antennas.)

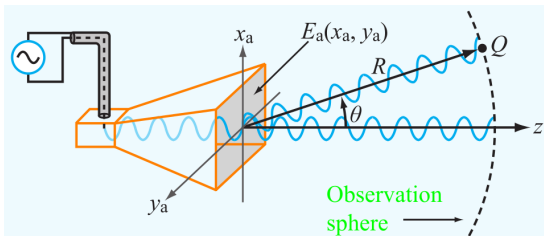


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

One application is microwave communication systems where

- high directivity is important
- $1 \text{ GHz} \leq f \leq 30 \text{ GHz} \iff 30 \text{ cm} \geq \lambda \geq 1 \text{ cm}$

Radiation from Large-Aperture Antennas (Continued)

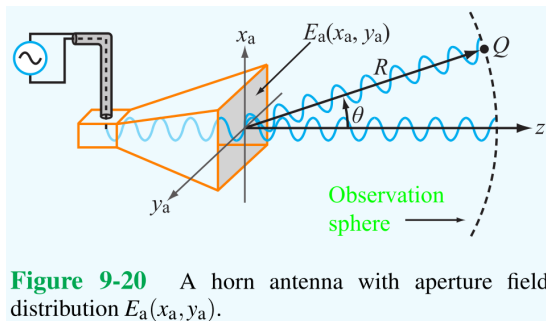


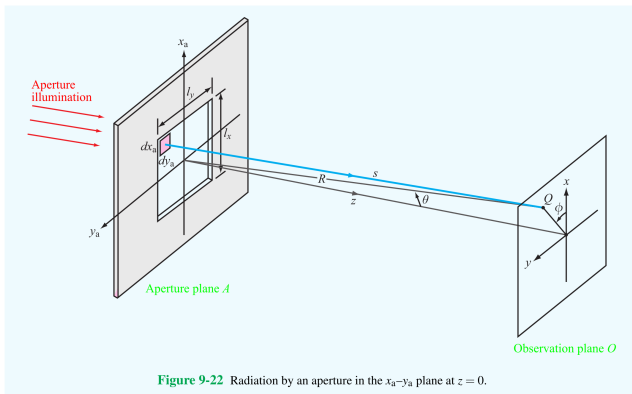
Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

Define a **2D coordinate system** (x_a, y_a) with its origin at the center of the aperture. The (scalar) **electric field** $E_a(x_a, y_a)$ **at the aperture** determines the radiation that the antenna emits. The electric field $E_a(x_a, y_a)$ is called the

- **electric-field aperture distribution**
- or **illumination**.

Radiation from Large-Aperture Antennas (Continued)

More generally, assume that an antenna has a **rectangular aperture** with length ℓ_x in the x_a direction and length ℓ_y in the y_a direction.

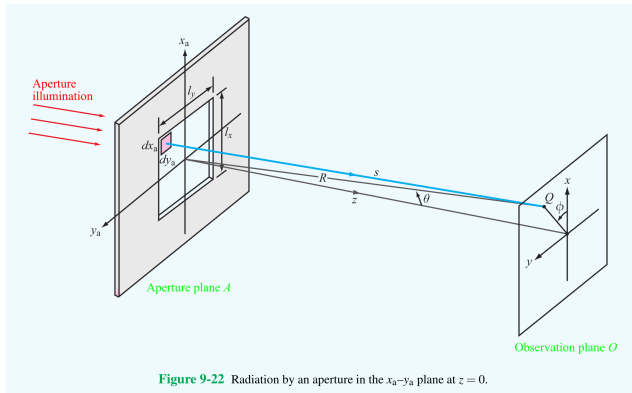


The goal is to determine the electric field at a point Q on an “observation” plane that is parallel to the aperture.

Radiation from Large-Aperture Antennas (Continued)

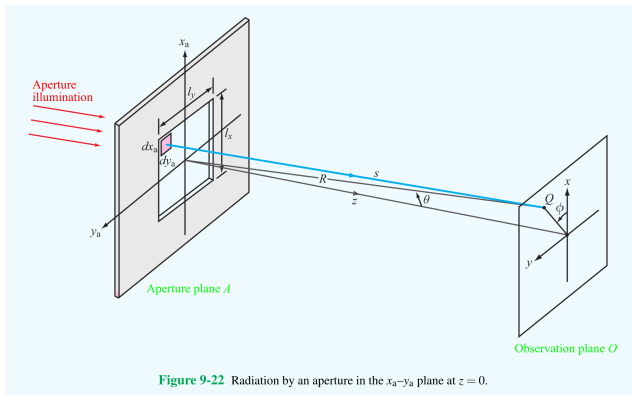
For the observation plane,

- the origin lies on the z axis of the aperture,
- the z axis lies along the z axis of the aperture,
- the x and y axes are parallel to the x_a and y_a axes, respectively,
- and every point Q on the plane is in the far field of the aperture.



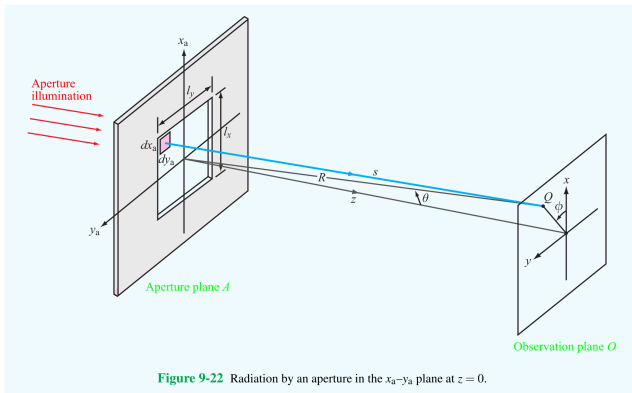
Radiation from Large-Aperture Antennas (Continued)

Let (R, θ, ϕ) be the spherical (aperture) coordinates of a point Q on the observation plane. Here, the **elevation angle** θ is the angle from the positive z axis and ϕ is the angle from the positive x_a axis (or x axis).



Note: These coordinate systems differ from the coordinate system used to analyze current source antennas. (Be careful when comparing formulas.)

Radiation from Large-Aperture Antennas (Continued)



For Q to be in the far field of the aperture,

$$R \geq 2d^2/\lambda, \quad (35)$$

where d is the **longest linear dimension** of the aperture.

Radiation from Large-Aperture Antennas (Continued)

Let $\tilde{E}(R, \theta, \phi)$ be the **phasor of the electric field at Q** and $\tilde{E}_a(x_a, y_a)$ be the **phasor of the illumination**. Then **Kirchhoff's scalar diffraction theory** implies

$$\tilde{E}(R, \theta, \phi) = \frac{j}{\lambda} \left(\frac{e^{-jkR}}{R} \right) \tilde{h}(\theta, \phi), \quad (36)$$

where

$$\tilde{h}(\theta, \phi) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \tilde{E}_a(x_a, y_a) e^{jk \sin \theta (x_a \cos \phi + y_a \sin \phi)} dx_a dy_a \quad (37)$$

is the **form factor** and e^{-jkR}/R is the **spherical propagation factor**.

- The integration is done over the aperture (despite the infinite limits).
- The form factor adds the contributions from the isotropic sources across the aperture.
- The spherical propagation factor represents the attenuation of the electric field as a function of R .

Radiation from Large-Aperture Antennas (Continued)

Assuming the unbounded medium is **lossless** and has the **EM properties of free space**, the **power density at point Q** is

$$S(R, \theta, \phi) = \frac{|\tilde{E}(R, \theta, \phi)|^2}{2\eta_0} = \frac{|\tilde{h}(\theta, \phi)|^2}{2\eta_0 \lambda^2 R^2}. \quad (38)$$

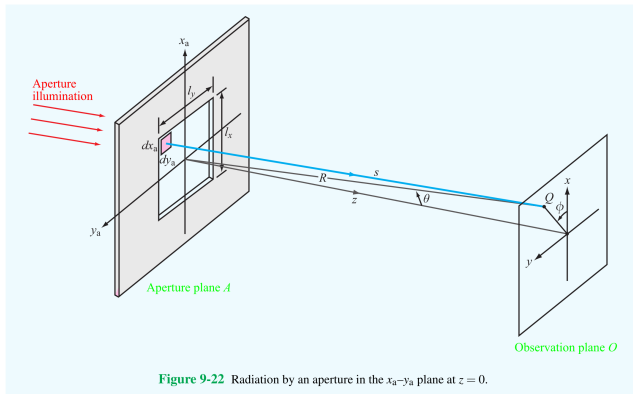
Though not justified, the **polarization of $\tilde{E}(R, \theta, \phi)$ is the same as the polarization of $\tilde{E}_a(x_a, y_a)$** .

Rectangular Aperture with Uniform Illumination

Suppose that the **illumination is uniform**. This means

$$\tilde{E}_a(x_a, y_a) = \begin{cases} E_0, & -\ell_x/2 \leq x_a \leq \ell_x/2, \quad -\ell_y/2 \leq y_a \leq \ell_y/2 \\ 0, & \text{otherwise} \end{cases}, \quad (39)$$

where E_0 is a **constant**. Also, suppose $\phi = 0$ so Q lies on the $+x$ axis.



Rectangular Aperture with Uniform Illumination (Cont.)

Under these circumstances,

$$\begin{aligned}\tilde{h}(\theta, \phi) &= \tilde{h}(\theta) = \int_{-\ell_y/2}^{\ell_y/2} \int_{-\ell_x/2}^{\ell_x/2} E_0 e^{jk \sin \theta (x_a \cos(0) + y_a \sin(0))} dx_a dy_a \\ &= E_0 \int_{-\ell_y/2}^{\ell_y/2} \int_{-\ell_x/2}^{\ell_x/2} e^{jk x_a \sin \theta} dx_a dy_a.\end{aligned}\quad (40)$$

Recall that the wavenumber $k = 2\pi/\lambda$ and define the constant

$$u = k \sin \theta = \frac{2\pi}{\lambda} \sin \theta. \quad (41)$$

Then

$$\begin{aligned}\tilde{h}(\theta) &= E_0 \int_{-\ell_y/2}^{\ell_y/2} \int_{-\ell_x/2}^{\ell_x/2} e^{jux_a} dx_a dy_a = \frac{2E_0\ell_y}{u} \sin(u\ell_x/2) \\ &= E_0 A_p \frac{\sin(\pi\ell_x \sin \theta/\lambda)}{\pi\ell_x \sin \theta/\lambda} = E_0 A_p \text{sinc}(\pi\ell_x \sin \theta/\lambda).\end{aligned}\quad (42)$$

Rectangular Aperture with Uniform Illumination (Cont.)

Here,

$$A_p = \ell_x \ell_y \quad (43)$$

is the area of the aperture and the sinc function is defined as

$$\text{sinc}(t) = \sin(t)/t. \quad (44)$$

Altogether, the **phasor of the electric field at Q** is

$$\tilde{E}(R, \theta, \phi) = \frac{j}{\lambda} \left(\frac{e^{-jkR}}{R} \right) \tilde{h}(\theta) = \frac{jE_0 A_p}{\lambda} \left(\frac{e^{-jkR}}{R} \right) \text{sinc}(\pi \ell_x \sin \theta / \lambda) \quad (45)$$

and the **power density at Q** is

$$S(R, \theta) = S_0 \text{sinc}^2(\pi \ell_x \sin \theta / \lambda), \quad (46)$$

where

$$S_0 = S(R, 0) = \frac{E_0^2 A_p^2}{2\eta_0 \lambda^2 R^2} \quad (47)$$

is the maximum power density.

Rectangular Aperture with Uniform Illumination (Cont.)

Antenna Pattern

The **normalized radiation intensity** is

$$\begin{aligned} F(\theta) &= \frac{S(R, \theta)}{S_{\max}} = \frac{S(R, \theta)}{S_0} \\ &= \text{sinc}^2(\pi \ell_x \sin \theta / \lambda) \\ &= \text{sinc}^2(\pi \gamma), \end{aligned} \quad (48)$$

where

$$\gamma = \frac{\ell_x}{\lambda} \sin \theta. \quad (49)$$

Note that γ is **not** the propagation constant.

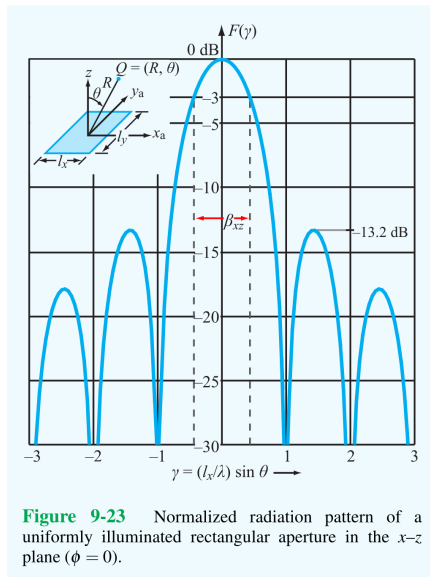


Figure 9-23 Normalized radiation pattern of a uniformly illuminated rectangular aperture in the x - z plane ($\phi = 0$).

Rectangular Aperture with Uniform Illumination (Cont.)

Beamwidth

The beamwidth can be found using the normalized radiation intensity $F(\theta)$. The **half-power beamwidth in the x - z plane** is

$$\beta_{xz} \approx 0.88 \frac{\lambda}{\ell_x} \text{ (rad)} \quad (50)$$

and the **half-power beamwidth in the y - z plane** is

$$\beta_{yz} \approx 0.88 \frac{\lambda}{\ell_y}. \quad (51)$$

The **uniform illumination** yields the **narrowest beamwidth**.

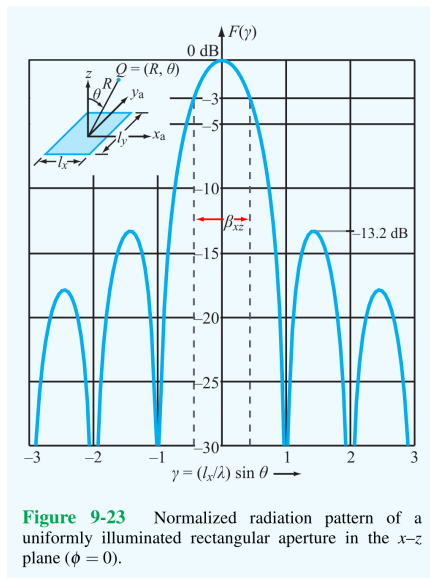


Figure 9-23 Normalized radiation pattern of a uniformly illuminated rectangular aperture in the x - z plane ($\phi = 0$).

Rectangular Aperture with Uniform Illumination (Cont.)

Applying a **taper**, which decreases $E_a(x_a, y_a)$ away from the center of the aperture, will decrease the side lobes at the expense of widening the main lobe. (This is like applying a window in signal processing.)

Directivity

The **directivity** of the antenna is

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \approx \frac{4\pi\ell_x\ell_y}{\lambda^2} = \frac{4\pi A_p}{\lambda^2}. \quad (52)$$

The **effective area** A_e of an antenna is defined using its directivity, specifically

$$A_e = \frac{\lambda^2}{4\pi} D. \quad (53)$$

So, for this type of antenna,

$$A_p \approx A_e. \quad (54)$$

Antenna Arrays

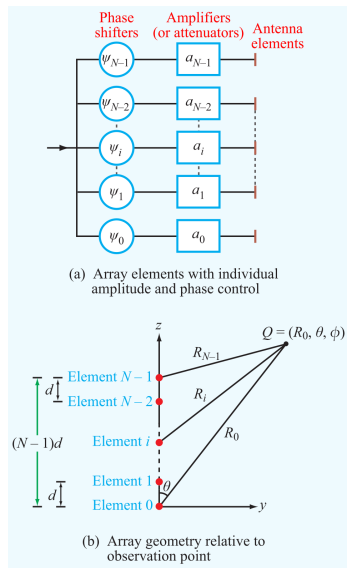
When **two or more antennas are combined**, the direction and shape of the **overall antenna pattern can be controlled**, even when the individual antennas are very simple. In this case, the group of antennas is called an **antenna array** and operates as a single antenna. The antenna pattern of an array is controlled through the **amplitudes and relative phases of signals emitted** through the antennas. Some applications of this are

- electronic steering
- multiple-beam generation
- beam forming

Linear Antenna Arrays

Consider a **linear array of N identical transmitting antennas distributed along the z -axis.**

- All of the antennas are **fed by a single oscillator** through a branching network.
- The **phase** and **amplitude** of the signal fed to each antenna is set by a **phase shifter** and an **amplifier** in the branch for the antenna, respectively.
- An antenna is an **element** of the array.
- Antenna/Element 0 is at the **origin**.
- Antenna/Element i is **located at** $(x_i, y_i, z_i) = (0, 0, id)$ for a **constant distance d** and $i = 0, \dots, N - 1$.



Linear Antenna Arrays (Continued)

The **phasor of the electric field intensity** in the far field for each element of the array has the form

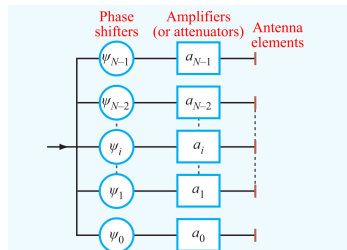
$$\tilde{E}_e(R, \theta, \phi) = \frac{e^{-jkR}}{R} \tilde{f}_e(\theta, \phi), \quad (55)$$

where

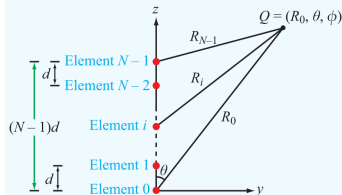
- the **spherical propagation factor** is

$$e^{-jkR} / R \quad (56)$$

- and $\tilde{f}_e(\theta, \phi)$ describes the **dependence of $\tilde{E}_e$ on direction**.



(a) Array elements with individual amplitude and phase control

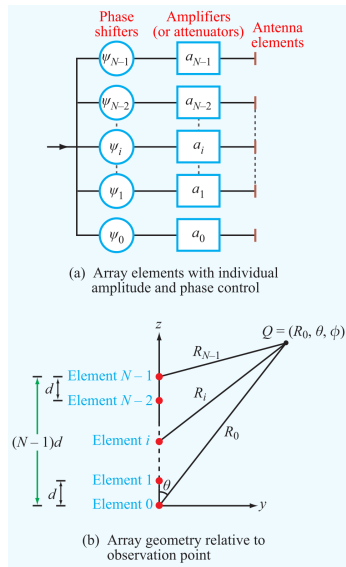


(b) Array geometry relative to observation point

Linear Antenna Arrays (Continued)

Assuming that the array is transmitting into a medium with the **EM properties of free space**, such as air, the **average power density** due to $\tilde{E}_e$ is

$$\begin{aligned} S_e(R, \theta, \phi) &= \frac{1}{2\eta_0} |\tilde{E}_e(R, \theta, \phi)|^2 \\ &= \frac{1}{2\eta_0 R^2} |\tilde{f}_e(\theta, \phi)|^2. \end{aligned} \quad (57)$$



Linear Antenna Arrays (Continued)

Now, suppose $Q = Q(R_0, \theta, \phi)$ is a point in the **far field of all of the antennas** and that R_i is the distance from antenna i to Q . Then the **phasor of the electric field due to antenna i at Q** is

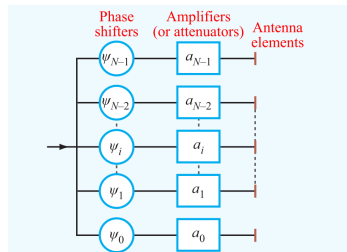
$$\tilde{E}_i(R_i, \theta, \phi) = A_i \frac{e^{-jkR_i}}{R_i} \tilde{f}_e(\theta, \phi), \quad (58)$$

where

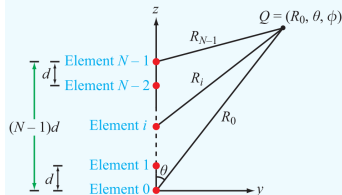
$$A_i = a_i e^{j\psi_i} \quad (59)$$

is the **complex feeding coefficient** for antenna i . Here,

- the i^{th} phase shifter sets ψ_i
- and the i^{th} amplifier sets a_i .



(a) Array elements with individual amplitude and phase control



(b) Array geometry relative to observation point

Linear Antenna Arrays (Continued)

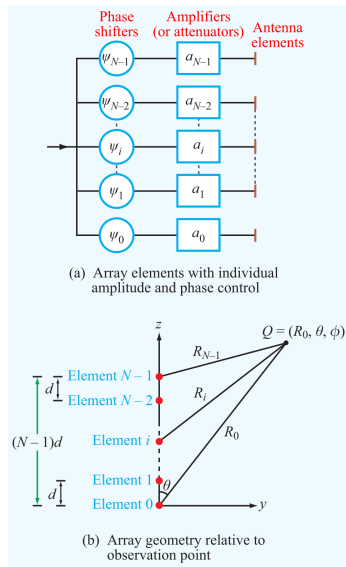
The **length of the array** is

$$\ell = (N - 1)d. \quad (60)$$

The requirement

$$R_0 \geq \frac{2\ell^2}{\lambda} = \frac{2(N - 1)^2 d^2}{\lambda} \quad (61)$$

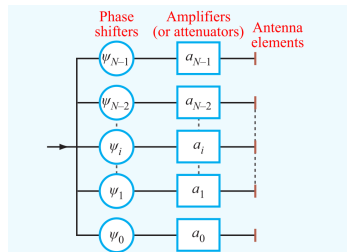
guarantees that $Q = Q(R_0, \theta, \phi)$ is **in the far field for all of the antennas**.



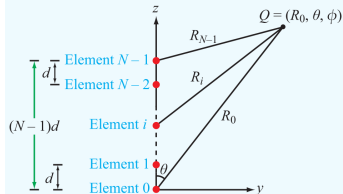
Linear Antenna Arrays (Continued)

The **total electric field phasor** at Q is

$$\begin{aligned}
 \tilde{E}(R_0, \theta, \phi) &= \sum_{i=0}^{N-1} \tilde{E}_i(R_i, \theta, \phi) \\
 &= \sum_{i=0}^{N-1} A_i \frac{e^{-jkR_i}}{R_i} \tilde{f}_e(\theta, \phi) \\
 &= \left(\sum_{i=0}^{N-1} A_i \frac{e^{-jkR_i}}{R_i} \right) \tilde{f}_e(\theta, \phi). \quad (62)
 \end{aligned}$$



(a) Array elements with individual amplitude and phase control



(b) Array geometry relative to observation point

Linear Antenna Arrays (Continued)

Since Q is in the far field of all of the antennas, it's possible to make 2 approximations in the expression for $\tilde{E}(R_0, \theta, \phi)$.

- 1 For R_i in the denominators of the spherical propagation factors,

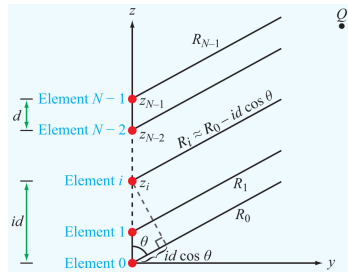
$$R_i \approx R_0 \quad (63)$$

- 2 Assume that the line segments connecting Antenna i to Q are parallel for all i . This leads to the approximation

$$e^{-jkR_i} \approx e^{-jk(R_0 - z_i)} = e^{-jk(R_0 - id \cos \theta)} = e^{-jkR_0} e^{jkid \cos \theta}. \quad (64)$$

Applying these approximations,

$$\tilde{E}(R_0, \theta, \phi) = \left(\sum_{i=0}^{N-1} A_i \frac{e^{-jkR_i}}{R_i} \right) \tilde{f}_e(\theta, \phi) = \tilde{f}_e(\theta, \phi) \left(\frac{e^{-jkR_0}}{R_0} \right) \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta}. \quad (65)$$



Homework

Read Chapter 9.

Homework 3 was due today.

Homework 4 is posted and due Friday, June 5 at 10 pm. I might add more problems from Chapter 9 to the assignment.

Exam 2 will be on Tuesday, June 9, from 1:30 – 3:20 pm.